Tutorial 2[1]

All problems below except problem 7 are of major conceptual importance.

Recall the definitions of the following: **linear combination, span, subspace, linear map**

1. Let A be an $r \times c$ matrix and x a vector in $\mathbb{R}^c$. Show that Ax is a linear combination of certain vectors obtained from entries of A.

2. Recall/check that the following are subspaces of $\mathbb{R}^c$.

   (i) For a subset S of $\mathbb{R}^c$, $\text{span}(S) =$ set of all linear combinations of vectors from S.

   (ii) The set of solutions of $Ax = 0$ for an $r \times c$ matrix A.

   (a) We have seen that RREF(A) allows you to write a subspace of the second type as a subspace of the first type by finding a magic set of vectors. Do this for the following system.

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 2 \\ 2 & 4 & 0 & 4 & 6 \\ 1 & 2 & 2 & 5 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

   (b) (*Do this later. It is optional for now.*) Show how to write a subspace of first type as a subspace of second type by finding a magic matrix. Setup: Let $S = \{v_1, \dots, v_k\}$ be a subset of $\mathbb{R}^n$. Show how to explicitly find a matrix A such that

$$\text{span}(S) = \{x \in \mathbb{R}^n \mid Ax = 0\}.$$

3. Recall that "General solution of a consistent system is a translate of a subspace", i.e., of the form $W + u$ where W is a subspace (of what?) and u a vector. Demonstrate this explicitly for the following system.

$$\begin{pmatrix} 1 & -1 & 2 & 3 \\ 2 & -2 & 5 & 4 \\ -1 & 1 & -1 & -5 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 5 \\ 11 \\ -4 \end{pmatrix}$$

4. For an $r \times c$ matrix, recall/check that the map $f(x) = Ax$ from $\mathbb{R}^c$ to $\mathbb{R}^r$ is linear, which means it preserves vector addition and scalar multiplication. Show that every linear map L from $\mathbb{R}^c$ to $\mathbb{R}^r$ comes in this fashion, i.e., for a given linear map L, there is a unique matrix A such that $L(x) = Ax$.

5. This exercise has many parts and it will take some time to do all of them, but several are within your reach now. Do what you can at the moment!

   When is $f(x) = Ax$ injective? Surjective? Give an "if and only if" criterion for each of the two conditions in terms of each of the following. (So the final answer will be a table with two columns and five rows, the two entries in the top row being injective and surjective.)

   (a) Solutions of a suitable linear system

   (b) REF(A)

   (c) Column vectors of A

   (d) Row vectors of A

[1] Gemini turned the text into LaTeX and supplied the two numerical examples.

6. (a) Show that doing an elementary row operation on a matrix A results in a matrix EA, where E is a suitable matrix, called an **elementary matrix**. Describe all three types of elementary matrices. (b) Show that E itself can be obtained using an elementary row operation on a very simple matrix.

7. This problem is useful as problem-solving/proof writing practice, but not conceptually central. Recall the following:

   (i) A matrix is said to be in REF if leading nonzero entries in successive rows move strictly to the right and all 0 rows are at the bottom. We can convert any matrix to REF by doing a sequence of row operations by doing a forward/downward pass. We also scale the leading nonzero entries, called pivots, to 1.

   (ii) A matrix is said to be in RREF if it is in REF and moreover all entries above any pivot are 0. We can convert any REF matrix to RREF by a backward/upward pass, by killing entries above pivots column by column by starting from the right.

   (a) Show that starting with an REF matrix, we can get to an RREF matrix by treating columns in any order.

   (b) Show that any matrix can be converted to a unique matrix in RREF by row operations.